Md. Biplob Hosen — Curriculum Vitae

MSc in Water and Environmental Management | GIS, Remote Sensing and Hydrological Modelling
Institute of Water & Environment, Dhaka University of Engineering & Technology (DUET), Gazipur-1707, Bangladesh
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Professional Summary

Water and environmental researcher with an MSc in Water and Environmental Management (CGPA 3.92/4.00) from DUET. Co-author of nine peer-reviewed journal articles and seventeen international conference papers on GIS- and remote-sensing-based assessment of groundwater potential, flood and landslide susceptibility, soil erosion, land surface temperature, water quality and site suitability, using AHP, Fuzzy AHP, artificial neural networks and hydrological and hydraulic models (SWAT, HEC-HMS, HEC-RAS). Research Assistant on two research projects at DUET (July 2024 – June 2026) and recipient of two presenting-paper awards at international conferences. Seeking a PhD position in climate-resilient water resources planning, flood risk reduction, and spatial environmental analysis.

Research Interests

- GIS and remote sensing in hydrology, watershed management and urban mapping
- Hydrological and rainfall–runoff modelling for water resources management
- Climate change impact assessment on water resources
- Groundwater recharge and potential zone mapping; soil erosion modelling
- Flood susceptibility, vulnerability and risk mapping; geomorphology and hydrology
- Artificial intelligence and multi-criteria decision-making (AHP, Fuzzy AHP, ANN) in water resources management
- Nature-based solutions for disaster and climate resilience
- Ecosystem services and green infrastructure planning

Education

M.Sc. in Water and Environmental Management | 2023 – 2025

Institute of Water & Environment, Dhaka University of Engineering & Technology (DUET), Gazipur-1707, Bangladesh

CGPA: 3.92 / 4.00

Thesis: *Assessment of Land Use and Climate Change Impacts on Streamflow in the Sangu River*
(Supervisor: Md. Rabiul Islam, Assistant Professor, Institute of Water & Environment, DUET)

Relevant coursework: Remote Sensing and GIS Applications in Water and Environmental Management; Integrated Water Management; Environmental Hydrology and Water Resources; Water, Energy and Environment; Coastal Risk Management; Solid Waste Management

B.Sc. (Honors) in Environmental Science and Disaster Management | 2018 – 2021

Gopalganj Science and Technology University, Gopalganj, Bangladesh

CGPA: 3.01 / 4.00

Research Experience

Research Assistant | July 2024 – June 2026

Institute of Water & Environment, DUET, Gazipur-1707, Bangladesh

- *Application of AHP, Fuzzy-AHP and ANN in multi-criteria decision-making to select optimal locations for electric vehicle charging stations (EVCSs) in Dhaka District, Bangladesh* (2024 – 2025). Project Director: Md. Rabiul Islam, Assistant Professor, Institute of Water & Environment, DUET.
- *Towards Sustainable Water Management: Assessing Water Security in the Shitalakshya River Basin through Multi-Temporal Blue and Green Water Footprints* (2025 – 2026). Project Director: Dr. Mohammad Nazim Uddin, Professor, Department of Civil Engineering, DUET. Associate Director: Md. Rabiul Islam, Assistant Professor, Institute of Water & Environment, DUET.

Peer-Reviewed Journal Articles

1. Kundu, S., **Hosen, M. B.**, Popy, S., Tahera-Tun-Humayra, U., Rahman, M. M., Salauddin, M., et al. (2026). Application of the fuzzy analytical hierarchy process for flood vulnerability assessment in Sylhet District, Bangladesh. *Discover Environment, 4*(1), 588.
2. Kader, Z., Islam, M. R., Miah, M., Aziz, M. T., Haque, A., Tumon, M. S. H., Mamun, M. A. A., **Hosen, M. B.**, Tahera-Tun-Humayra, U., Hasan, M., & Akanda, A. S. (2026). Performance assessment of multicriteria decision making techniques and artificial neural network to predict future groundwater potential zones. *Ecological Indicators, 183*, 114619. https://doi.org/10.1016/j.ecolind.2026.114619 [Q1; IF 7.7]
3. **Hosen, M. B.**, Islam, M. R., Tahera-Tun-Humayra, U., Sharker, R., Kader, Z., Aziz, M. T., Miah, M., Hasan, M., Pervin, R., Hossain, M. A., & Tofiquzzaman, M. (2025). Assessing land suitability for dragon fruit cultivation in Bangladesh: A GIS-based AHP approach. *Smart Agricultural Technology, 12*, 101241. https://doi.org/10.1016/j.atech.2025.101241 [Q1; IF 5.7]
4. Islam, M. R., **Hosen, M. B.**, Hossain, M. M., Ayejoto, D. A., & Khan, M. Y. A. (2025). Future projections of landslide susceptibility in the Hill Tracts of Bangladesh: An AHP-based approach. *Earth Systems and Environment, 9*(4), 3017–3041. https://doi.org/10.1007/s41748-025-00776-y [Q1; IF 4.7]
5. Aziz, M. T., Haque, A., Islam, M. R., **Hosen, M. B.**, Kader, Z., Aziz, M. A., & Saha, O. R. (2025). Site suitability assessment for solar-powered green hydrogen production plants: A GIS-based AHP and Fuzzy AHP approach for Bangladesh. *Renewable Energy, 254*, 123675. https://doi.org/10.1016/j.renene.2025.123675 [Q1; IF 9.0]
6. Tahera-Tun-Humayra, U., Islam, M. R., **Hosen, M. B.**, Kader, Z., Sharker, R., Hasan, M., Aziz, M. T., Miah, M., & Pervin, R. (2025). Groundwater potential zone mapping using analytical hierarchy process (AHP) and GIS for Narshingdi District, Bangladesh. *Environmental Challenges, 21*, 101335. https://doi.org/10.1016/j.envc.2025.101335 [Q1; CiteScore 11.0]
7. Sharker, R., Islam, M. R., **Hosen, M. B.**, Kader, Z., Aziz, M. T., Tahera-Tun-Humayra, U., Hossain, M. A., Pervin, R., Hasan, M., & Roy, A. (2025). GIS-based AHP approach to flood susceptibility assessment in Tangail district, Bangladesh. *Journal of Earth System Science, 134*(1), 26. https://doi.org/10.1007/s12040-024-02480-3 [Q2; IF 1.3]
8. Miah, M., Islam, M. R., Imran, H. M., Kader, Z., Aziz, M. T., Mamun, M. A. A., **Hosen, M. B.**, Islam, M. R., Tahera-Tun-Humayra, U., & Hasan, M. (2025). Assessing soil erosion in the

Ghorautra River Basin of Bangladesh: Using SCS-CN and RUSLE methods within the GIS-RS framework. *Journal of the Indian Society of Remote Sensing*, *53*(12), 4383–4403. https://doi.org/10.1007/s12524-025-02276-y [Q2; IF 2.2]

9. Bhowmik, D., Kader, Z., **Hosen, M. B.**, & Hossain, S. (2025). Exploring socioeconomic vulnerability and natural disasters impacts on water access in Southwestern Bangladesh. *DYSONA – Applied Science*, *6*(1), 186–199. https://doi.org/10.30493/das.2024.483646

Manuscripts Under Review and In Preparation

Under review

1. Tahera-Tun-Humayra, U., Islam, M. R., **Hosen, M. B.**, Sharker, R., Aziz, M. T., Kader, Z., Miah, M., Hasan, M., & Pervin, R. (2025). Projected land surface temperature response to land use land cover indices in Dhaka District, Bangladesh. Under review, *Climate Services*.
2. Mona, M. G., Tahera-Tun-Humayra, U., & **Hosen, M. B.** (2025). Hydro-chemical analysis of groundwater and river water for drinking in Paikgachha Upazila of Khulna District, Bangladesh. Under review, *Physics and Chemistry of the Earth, Parts A/B/C*.
3. Islam, M. R., Tahera-Tun-Humayra, U., Alam, M. A., **Hosen, M. B.**, & Aziz, M. T. (2025). Strategic planning of urban green spaces for mitigating the urban heat island effect: A threshold-based perspective. Under review, *City and Environment Interactions*.
4. Sharker, R., Islam, M. R., **Hosen, M. B.**, Tahera-Tun-Humayra, U., Aziz, M. T., Kader, Z., Hasan, M., Pervin, R., & Hossain, M. A. (2025). Spatiotemporal change dynamics in land use and land cover, land surface temperature, and urban heat island in Gazipur District, Bangladesh. Under review, *Computational Urban Science*.
5. Sharker, R., Islam, M. R., **Hosen, M. B.**, Kader, Z., Tahera-Tun-Humayra, U., Miah, M., Aziz, M. T., Pervin, R., & Hasan, M. (2025). AHP-based multi-hazard (flood, cyclone, and landslide) susceptibility assessment of Halda River Basin, Bangladesh. Under review, *Climate Risk Management*.
6. Miah, M., Islam, M. R., Kader, Z., Aziz, M. T., **Hosen, M. B.**, Tahera-Tun-Humayra, U., Haque, A., Pervin, R., Hossain, M. A., & Hasan, M. (2025). Delineating groundwater recharge potential zones: A geoinformatics-based spatial analysis. Under review, *Journal of Geosystem and Geoenvironment*.

In preparation

- Spatiotemporal assessment of meteorological drought using SPEI-12 in Barisal and Khulna divisions, Bangladesh.

Conference Papers

Seventeen papers presented at international conferences, newest first. Full conference names are listed under Awards and Conferences Attended.

1. **Hosen, M. B.**, Pervin, R., Islam, M. R., …, & Kader, Z. (2026). Spatiotemporal assessment of LST and LULC variability during summer in Narayanganj District, Bangladesh. *ICSWER 2026*, April 2026.
2. Tahera-Tun-Humayra, U., Pervin, R., **Hosen, M. B.**, Islam, M. R., & Fatama, M. K. (2026). City-specific cooling thresholds of urban green and blue spaces for urban heat island mitigation in Bangladesh. *ICSWER 2026*, April 2026.

3. Hossain, M. A., Pervin, R., Islam, M. R., …, & Tofiquzzaman, M. (2026). Satellite-based assessment of air quality dynamics in the industrial city of Gazipur, Bangladesh. *ICSWER 2026*, April 2026.
4. **Hosen, M. B.**, Tofiquzzaman, M., Islam, M. R., & Tahera-Tun-Humayra, U. (2026). Assessing flood vulnerability in Sylhet District with an AHP-based GIS approach. *8th International Conference on Civil Engineering for Sustainable Development*, February 2026.
5. Sharker, R., Tahera-Tun-Humayra, U., Pervin, R., Islam, M. R., **Hosen, M. B.**, & Fatama, M. K. (2026). Spatial analysis of multi-hazard risks in the Halda River Basin through AHP-based modeling. *8th International Conference on Civil Engineering for Sustainable Development*, February 2026.
6. Aziz, M. T., Pervin, R., Islam, M. R., …, & **Hosen, M. B.** (2025). Assessing the future flood hazard in the Matamuhuri River Basin using HEC-HMS and HEC-RAS. *International Conference on Civil Engineering Research & Innovations*, 12–14 December 2025, RUET, Rajshahi, Bangladesh.
7. Kader, Z., Pervin, R., Islam, M. R., Hossain, M. R., Hoque, J., Miah, M., **Hosen, M. B.**, Aziz, M. T., & Hasan, M. (2025). Assessment of runoff and potential soil erosion in the Dakatia river basin, Bangladesh. *International Conference on Civil Engineering Research & Innovations*, 12–14 December 2025, RUET, Rajshahi, Bangladesh.
8. **Hosen, M. B.**, Tofiquzzaman, M., Islam, M. R., & Tahera-Tun-Humayra, U. (2025). Evaluating drinking water quality and health risks in communities adjacent to the Modhumoti River, Bangladesh. *ICSHSD 2025*, October 2025, DUET, Gazipur, Bangladesh.
9. Tahera-Tun-Humayra, U., Hossain, M. A., Islam, M. R., & **Hosen, M. B.** (2025). Hydro-chemical assessment of groundwater and river water quality for drinking in Paikgachha Upazila, Khulna District, Bangladesh. *ICSHSD 2025*, October 2025, DUET, Gazipur, Bangladesh.
10. Roy, A., Pervin, R., Islam, M. R., Kader, Z., Sharker, R., **Hosen, M. B.**, & Tahera-Tun-Humayra, U. (2025). Assessment of heavy metal concentration and the physicochemical parameters of the outlet wastewater: A case study of the industrial zone in Gazipur District, Bangladesh. *ICSHSD 2025*, October 2025, DUET, Gazipur, Bangladesh.
11. Hasan, M., Islam, M. R., Pervin, R., Kader, Z., Aziz, M. T., **Hosen, M. B.**, Miah, M., Tahera-Tun-Humayra, U., Sharker, R., & Roy, A. (2024). A comparative study of AHP and Fuzzy-AHP approach in selecting solid waste disposal site in Barishal Division, Bangladesh. *IICASD 2024*, 7–9 November 2024, DUET, Gazipur, Bangladesh.
12. Ahmed, M. S., Islam, M. R., Haque, A., …, & **Hosen, M. B.** (2024). Shoreline change assessment using DSAS technique: A case study on the coast of Kuakata. *IICASD 2024*, 7–9 November 2024, DUET, Gazipur, Bangladesh.
13. Tahera-Tun-Humayra, U., Islam, M. R., Hossain, M. A., **Hosen, M. B.**, Aziz, M. T., Miah, M., Kader, Z., & Hasan, M. (2024). Measuring land surface temperature (LST) response to LULC indices in Dhaka District, Bangladesh. *IICASD 2024*, 7–9 November 2024, DUET, Gazipur, Bangladesh.
14. Miah, M., Islam, M. R., Hossain, M. A., **Hosen, M. B.**, Kader, Z., Aziz, M. T., Tahera-Tun-Humayra, U., & Hasan, M. (2024). Assessment of temporal changes in soil erosion at the Halda River Basin, Bangladesh. *IICASD 2024*, 7–9 November 2024, DUET, Gazipur, Bangladesh.
15. **Hosen, M. B.**, Islam, M. R., Islam, M. M., Akter, S., Tahera-Tun-Humayra, U., Hasan, M., Roy, A., & Aziz, M. T. (2024). Evaluating the saline water impact on agriculture in Gobra Union, Gopalganj. *2nd International Conference on Environment*, 2–3 March 2024, Khulna University, Khulna, Bangladesh.

16. Tahera-Tun-Humayra, U., Islam, M. R., Saliha, S., Mohammad, N., Pobittro, R., **Hosen, M. B.**, & Hasan, M. (2024). An evaluation of socio-economic condition and water quality parameters of Kopotokkho River. *2nd International Conference on Environment*, 2–3 March 2024, Khulna University, Khulna, Bangladesh.
17. Hasan, M., Islam, M. R., Parven, R., Hossain, M. A., **Hosen, M. B.**, Tahera-Tun-Humayra, U., Sharker, R., & Kader, Z. (2024). Groundwater potential zone mapping using analytical hierarchy process (AHP) and GIS for Narayanganj District, Bangladesh. *2nd International Conference on Environment*, 2–3 March 2024, Khulna University, Khulna, Bangladesh.

Awards and Conferences Attended

Awards

- **October 2025** | Presenting Paper Award, 1st International Conference on Science and Humanities for Sustainable Development (ICSHSD 2025), DUET, Gazipur, Bangladesh
- **7–9 November 2024** | Presenting Paper Award, 1st International Conference on Recent Innovation in Civil Engineering and Architecture for Sustainable Development (IICASD 2024), DUET, Gazipur, Bangladesh

Conferences attended and presented at

- **April 2026** | International Conference on Soil, Water and Environmental Resilience in the Coastal Zone of Bangladesh under a Changing Climate (ICSWER 2026) — 3 papers
- **February 2026** | 8th International Conference on Civil Engineering for Sustainable Development — 2 papers
- **12–14 December 2025** | International Conference on Civil Engineering Research & Innovations, Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh — 2 papers
- **October 2025** | 1st International Conference on Science and Humanities for Sustainable Development (ICSHSD 2025), DUET, Gazipur, Bangladesh — 3 papers
- **7–9 November 2024** | 1st International Conference on Recent Innovation in Civil Engineering and Architecture for Sustainable Development (IICASD 2024), DUET, Gazipur, Bangladesh — 4 papers
- **2–3 March 2024** | 2nd International Conference on Environment: Time for Nature and Natural Resource Management, Khulna University, Khulna, Bangladesh — 3 papers

Technical Skills

- **GIS and remote sensing:** ArcGIS, ArcGIS Pro, QGIS, ERDAS Imagine
- **Hydrological and hydraulic modelling:** SWAT, HEC-HMS, HEC-RAS, Delft3D
- **Decision analysis and machine learning:** AHP, Fuzzy AHP, artificial neural networks (ANN), MATLAB
- **Statistics and field survey tools:** SPSS, KoboToolbox

Training

- **June 2026** | Basic Python (60-hour professional course), Institute of Information & Communication Technology (IICT), DUET, Gazipur
- **July 2023** | Numerical Modeling, Artificial Intelligence, and GIS Technology
- **20 September – 13 October 2022** | Application of GIS and Remote Sensing in Research

Memberships and Activities

- Advisor, DUET Photographic Society (2025 – 2026)
- Chief Instructor, DUET Photographic Society (2024 – 2025)
- Vice President, BSMRSTU Photographic Society (2019 – 2022)
- Member, Bangladesh Poribeshbid Society (BPS)
- Member, Research and Skill Development Society (RSDS)

Referees

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